

PHARMACOLOGY DRUG STUDY CHART - Pharmacology I Exam 2

Class of Drugs	Generic Names (3+ Drugs)	Mechanism of Action (MOA)	Indications	Contraindications	Adverse Effects	BLACK BOX WARNING / Special Indications or Contraindications
Ophthalmic macrolides Lecture 4 · slides 14-15, 17	erythromycin, azithromycin Only 2 named in the lecture.	Inhibit RNA (ribonucleic acid)-dependent protein synthesis at the 50S ribosome (block transpeptidation), so no protein is made and the bacteria die.	Bacterial conjunctivitis. Erythromycin ointment is the most common and is soothing on an inflamed eye; it is also used to prevent ophthalmia neonatorum. Azithromycin costs more but needs fewer doses.	None listed; no drug interactions.	Eye irritation; hypersensitivity.	—
Ophthalmic fluoroquinolones Lecture 4 · slides 21-22	ciprofloxacin, ofloxacin, levofloxacin, moxifloxacin, gatifloxacin	Inhibit DNA (deoxyribonucleic acid) gyrase and topoisomerase IV, so supercoiled DNA cannot relax and the DNA breaks (double-stranded breaks).	Conjunctivitis and corneal ulcers. Preferred for corneal ulcers and for Pseudomonas aeruginosa. Preferred for conjunctivitis in contact lens wearers once keratitis is ruled out. Expensive, with frequent dosing.	None listed. Caution: rule out keratitis first in a contact lens wearer; emerging resistance; no interactions when given in the eye.	Ocular irritation; white precipitate (ciprofloxacin, about 17%); unpleasant taste after instillation.	Preferred for corneal ulcers and Pseudomonas aeruginosa.
Ophthalmic aminoglycosides Lecture 4 · slide 23	gentamicin, tobramycin Only 2 named in the lecture.	Bind the 30S ribosomal subunit and interfere with bacterial protein synthesis.	Conjunctivitis (solutions every 2 to 4 hours; ointment 2 to 3 times a day).	None listed; no clinically significant interactions.	Ocular irritation; corneal ulceration; reactive keratoconjunctivitis (with several days of use).	Reactive keratoconjunctivitis after several days of use; corneal ulceration is also listed.
Other topical antibacterials (sulfonamide, polypeptide, folate antagonist plus polymyxin) Lecture 4 · slides 18-20	sulfacetamide, bacitracin, trimethoprim with polymyxin B	Sulfacetamide: blocks folic acid synthesis (antagonizes PABA (para-aminobenzoic acid)). Bacitracin: blocks cell wall synthesis. Trimethoprim: blocks folic acid reduction; polymyxin B: binds membrane phospholipids, so cell contents leak.	Bacterial conjunctivitis (all three). Polytrim (trimethoprim with polymyxin B) is a solution; sulfacetamide comes as ointment or solution; bacitracin as an ointment.	Sulfacetamide: avoid in patients with a sulfonamide allergy. Bacitracin and Polytrim: no clinically significant interactions.	Ocular irritation (all three); sulfacetamide also causes allergic reactions.	Sulfacetamide: avoid in any patient with a sulfonamide allergy.
Antivirals for herpes simplex and herpes zoster eye disease Lecture 4 · slides 25-27	trifluridine, acyclovir, valacyclovir, famciclovir	Trifluridine inhibits thymidylate synthetase and is built into viral DNA (deoxyribonucleic acid) in place of thymidine, so viral replication stops. Mechanism not given for the other three.	Trifluridine (topical): herpes simplex keratitis and keratoconjunctivitis. Acyclovir: herpes zoster ophthalmicus, herpes simplex iridocyclitis. Valacyclovir, famciclovir (oral): herpes simplex keratitis, herpes zoster ophthalmicus.	None listed. No antiviral is used for viral conjunctivitis caused by adenoviruses; it is self-limited.	Trifluridine: ocular irritation, punctate keratopathy (pinpoint corneal damage), hypersensitivity.	Adenoviral conjunctivitis gets no antiviral: treat symptoms only.
Antivirals for cytomegalovirus retinitis Lecture 4 · slides 26, 28	ganciclovir, valganciclovir, foscarnet, cidofovir	Ganciclovir competitively inhibits binding of deoxyguanosine triphosphate to DNA (deoxyribonucleic acid) polymerase, blocking viral DNA synthesis. Mechanism not given for the others.	Cytomegalovirus retinitis: ganciclovir (intravenous, oral or intravitreal implant), valganciclovir (oral), foscarnet (intravenous or intravitreal), cidofovir (intravenous). Ganciclovir (Zirgan): herpetic keratitis.	None listed.	Ganciclovir (Zirgan): ocular irritation, punctate keratitis.	BLACK BOX Systemic ganciclovir, valganciclovir, cidofovir and foscarnet carry an FDA (Food and Drug Administration) boxed warning (bone marrow suppression, kidney injury, seizures with foscarnet); not stated in the lecture.

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Ocular antifungals (polyenes and imidazoles) Lecture 4 · slides 30-32	natamycin, amphotericin B, fluconazole, itraconazole, ketoconazole, miconazole	Natamycin binds sterols and raises fungal cell membrane permeability. Mechanism not given for amphotericin B or the imidazoles.	Natamycin (the only commercially available ophthalmic antifungal): conjunctivitis, keratitis (Aspergillus, Candida, Fusarium and others). Amphotericin B and the imidazoles: yeast and fungal keratitis, endophthalmitis.	None listed.	Natamycin: ocular irritation.	BLACK BOX Oral ketoconazole carries an FDA (Food and Drug Administration) boxed warning (liver injury; QT prolongation with interacting drugs); systemic itraconazole and amphotericin B also carry boxed warnings; not stated on the ocular antifungal slide. Risk factors for fungal eye infection: trauma, chronic surface disease, contact lenses, immunosuppression (including topical steroids).
Ophthalmic antihistamines (histamine-1 receptor antagonists) Lecture 4 · slides 39, 42-44	azelastine, alcaftadine, bepotastine, emedastine, epinastine, ketotifen, olopatadine	Inverse agonists at the histamine-1 (H1) receptor, still competitive with histamine: decrease capillary dilation, itch and swelling.	Ocular allergy. Dosed 1 to 2 times a day; onset within minutes; allow two weeks to judge full effect. Typically preferred over mast cell stabilizers.	None listed; no significant drug interactions.	Ocular irritation; headache; increased ocular dryness.	Preferred over mast cell stabilizers: onset within minutes.
Mast cell stabilizers Lecture 4 · slides 45-47	cromolyn, lodoxamide, nedocromil	Inhibit mast cell degranulation, limiting release of histamine, tryptase and prostaglandin D2; also dampen mediator release from basophils, eosinophils and neutrophils.	Predictable seasonal allergies in patients who do not tolerate other therapies. Full effect takes 5 to 14 days.	None listed. Limitation: not useful for acute symptoms (slow onset).	Ocular irritation; unpleasant taste; headache.	NOT for acute symptoms: 5 to 14 days to full effect, and often four-times-daily dosing.
Topical vasoconstrictors (imidazoline derivatives) Lecture 4 · slides 48-49	tetrahydrozoline, naphazoline, pheniramine with naphazoline	Vasoconstrictor that activates postjunctional alpha-adrenergic receptors on blood vessels, decreasing conjunctival edema (pheniramine adds an antihistamine). Systemically they target alpha-2 receptors.	Decreases conjunctival swelling (edema); over-the-counter; good for under 2 weeks.	Not for prolonged use. If no improvement in 72 hours, stop and see a provider (it could be something more serious). Accidental ingestion matters: systemically these act on alpha-2 receptors.	Rebound hyperemia after prolonged use and stopping.	Rebound hyperemia with prolonged use: limit to under 2 weeks (a key counseling point).
Ophthalmic NSAIDs (nonsteroidal anti-inflammatory drugs) Lecture 4 · slides 50-51	bromfenac, diclofenac, flurbiprofen, ketorolac, nepafenac	Block cyclooxygenase, so arachidonic acid is not converted to prostaglandins and thromboxanes (pro-inflammatory mediators).	Postoperative inflammation and pain; allergic conjunctivitis.	None listed. Not routinely recommended for conjunctivitis.	Lacrimation; keratitis; increased eye pressure; ocular irritation.	Not routinely recommended for conjunctivitis.
Ophthalmic glucocorticoids Lecture 4 · slides 52, 54-55	dexamethasone, prednisolone, difluprednate, fluorometholone, loteprednol, rimexolone, triamcinolone	Suppress the late-phase allergic reaction: inhibit phospholipase A2, so fewer arachidonic acid-derived inflammatory mediators; also inhibit fibrin and collagen deposition (less scarring).	Severe ocular allergy; anterior uveitis; external eye inflammatory disease; inflammation after ocular surgery. Given topically or intraocularly.	Generally reserved for refractory symptoms, limited to under 2 weeks. Cautions: infection, corneal ulcers, glaucoma and raised eye pressure, cataract, delayed wound healing.	Cataract; raised eye pressure (more with a family history) and glaucoma; infection (decreased immune function); delayed wound healing; corneal ulcers.	Limit to under 2 weeks. Soft steroids (fluorometholone, loteprednol, rimexolone) carry a lower risk of raised eye pressure.

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Tear substitutes (artificial tears) Lecture 4 · slides 56-57	balanced salt solution, carboxymethyl-cellulose, hydroxypropyl cellulose, polyvinyl alcohol	Hypotonic or isotonic solutions containing electrolytes, surfactants and thickeners; the thickeners increase time in the cul-de-sac.	Dry eye (alteration of the pre-corneal tear film and ocular surface). Many are over-the-counter.	None listed.	None listed.	Treat the underlying disease first (for example Sjogren syndrome, rheumatoid arthritis, vitamin A deficiency).
Ocular immunomodulator Lecture 4 · slide 58	cyclosporine Only 1 named in the lecture.	Inhibits production and release of interleukin-2, so less T cell activation: fewer inflammatory markers in the lacrimal gland and more tear production.	Chronic dry eye associated with inflammation (keratoconjunctivitis sicca); improves vision and comfort.	None listed; no drug interactions.	Ocular burning (17%); foreign body sensation; blurred vision.	—
Prostaglandin analogs (glaucoma) Lecture 4 · slides 64-66	latanoprost, travoprost, bimatoprost, tafluprost	Analogues of prostaglandin F ₂ -α: bind prostaglandin receptors and lower eye pressure by increasing aqueous outflow (exact mechanism unclear).	Glaucoma and ocular hypertension: first-line, the most commonly used glaucoma drugs, once-daily dosing.	None listed. Do not exceed once-daily dosing: more frequent dosing inhibits the pressure-lowering effect.	Conjunctival hyperemia; ocular irritation; changes in eyelash length and iris color. Limited systemic side effects.	Never dose more than once daily (it blunts the effect).
Ophthalmic beta blockers (glaucoma) Lecture 4 · slides 64, 67-68, 74	timolol, carteolol, levobunolol, betaxolol	Block beta receptors in the ciliary body epithelium: less catecholamine activation, less cyclic adenosine monophosphate, less aqueous humor production.	Glaucoma and ocular hypertension (second-line; decreases aqueous production). Start with a prostaglandin analogue or a beta blocker.	Avoid or use cautiously in heart failure, bradycardia, heart block and asthma. Beta-1 selective agents (betaxolol) carry less bronchoconstriction risk in asthma.	Worsening heart failure; bradycardia; heart block; increased airway resistance (asthma).	Nonselective agents (timolol, carteolol, levobunolol) work better but cause more side effects; betaxolol is beta-1 selective.
Alpha-2 adrenergic agonists (glaucoma) Lecture 4 · slides 64, 69	apraclonidine, brimonidine Only 2 named in the lecture.	Alpha-2 agonists: presynaptically decrease catecholamine release, and postsynaptic alpha-2 activation decreases aqueous production; aqueous outflow also increases.	Glaucoma and ocular hypertension (decrease aqueous production and increase outflow).	Contraindicated in children under 2 years (central nervous system depression and apnea).	Ocular irritation; hyperemia (rebound effect); pruritus; allergic conjunctivitis (less common with brimonidine).	Contraindicated in children under 2 years: central nervous system depression and apnea.
Carbonic anhydrase inhibitors (glaucoma) Lecture 4 · slides 64, 70, 73	dorzolamide, brinzolamide Only 2 named in the lecture.	Inhibit carbonic anhydrase in the ciliary body epithelium: less bicarbonate, less fluid transport, lower eye pressure (decreased aqueous production).	Glaucoma and ocular hypertension. Available in combinations (dorzolamide with timolol; brinzolamide with brimonidine) for fewer drops and better adherence.	None listed.	Bitter taste (25%); burning or stinging after administration (33%); allergic conjunctivitis.	—
Cholinergic agonists (miotics, glaucoma) Lecture 4 · slides 64, 72	acetylcholine, carbachol, pilocarpine	Activate muscarinic receptors, causing ciliary muscle contraction that facilitates aqueous outflow.	Glaucoma (increase aqueous outflow). Acetylcholine (Miochol-E) is used in surgical settings.	Younger patients are usually intolerant because of visual blurring; poor compliance from side effects and frequent administration.	Fixed small pupils; myopia; visual disturbances; headaches.	Poorly tolerated by younger patients (visual blurring).
Ocular anesthetics Lecture 4 · slide 75	tetracaine, proparacaine Only 2 named in the lecture.	Inhibit sodium influx into the neuron, preventing signal propagation.	Tonometry; foreign body removal; superficial corneal surgery.	None listed. Do not write prescriptions for these.	Hypersensitivity; burning sensation. Eyes stay numb for 10 to 20 minutes with no blink reflex.	No blink reflex for 10 to 20 minutes; do not write prescriptions for these.

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Mydriatics and cycloplegics (antimuscarinics and sympathomimetic) Lecture 4 · slides 76-77	atropine, cyclopentolate, tropicamide, phenylephrine	Antimuscarinics competitively block muscarinic acetylcholine receptors, causing mydriasis. Phenylephrine is an adrenergic receptor agonist that causes mydriasis and is more reactive to light.	Diagnostic use (fundoscopic examination) and uveitis (prevent synechiae, relieve ciliary spasm).	None listed.	Photosensitivity; blurred vision. Phenylephrine: photosensitivity, conjunctival hyperemia.	—
Diagnostic dye Lecture 4 · slide 78	fluorescein Only 1 named in the lecture.	Reveals epithelial defects of the cornea and conjunctiva (a stain).	Anterior segment staining; disclosing corneal injury.	None listed.	Hypersensitivity; burning sensation.	—
Penicillins (first-line for ear, sinus and throat infections) Lecture 5 · slides 5, 7-8	amoxicillin, amoxicillin-clavulanate, benzathine penicillin G	Not given in this lecture.	Otitis media: high-dose amoxicillin is the mainstay; amoxicillin-clavulanate if severe, resistant, or antibiotics in the past month. Rhinosinusitis: amoxicillin-clavulanate. Group A streptococcal pharyngitis: amoxicillin or one benzathine penicillin G injection.	Penicillin allergy: use an alternative (see the next row).	Not given in this lecture.	Use a rapid strep test before treating pharyngitis: most cases are viral.
Alternatives for penicillin allergy or failed therapy (cephalosporins, macrolide, lincosamide, fluoroquinolone) Lecture 5 · slides 5-8	cefdinir, cefixime, cephalexin, ceftriaxone, azithromycin, clindamycin, levofloxacin	Not given in this lecture.	Penicillin-allergic: otitis, cefdinir or azithromycin; sinusitis, clindamycin plus cefixime or levofloxacin; pharyngitis, cephalexin, clindamycin or azithromycin. Failed therapy (no better in 3 days; no antibiotics in past 3 months): amoxicillin-clavulanate or cefdinir; may escalate to ceftriaxone.	None listed. Up to 50% of Streptococcus pneumoniae are resistant to macrolides.	Not given in this lecture.	BLACK BOX Levofloxacin: FDA (Food and Drug Administration) boxed warning (tendon rupture, nerve damage, myasthenia gravis worsening). Clindamycin: boxed warning (Clostridioides difficile colitis). Not stated in the lecture. Ceftriaxone (intramuscular) is the step-up for failure.
Otic anti-infectives (with or without a glucocorticoid) Lecture 5 · slides 9-10	ciprofloxacin, ofloxacin, ciprofloxacin with dexamethasone, neomycin with polymyxin B and hydrocortisone	The anti-infective inhibits bacterial growth; the glucocorticoid decreases inflammatory cytokine production.	Otitis media and otitis externa. Ciprofloxacin with dexamethasone (Ciprodex) is expensive; ofloxacin with dexamethasone ophthalmic drops are the alternative.	Polymyxin B: not recommended with a ruptured eardrum or ear tubes in place (cochlear damage, hearing loss).	Neomycin: hypersensitivity. Polymyxin B: cochlear damage and hearing loss.	Neomycin with polymyxin B drops: avoid with a ruptured eardrum or ear tubes (inner ear damage).
Systemic imidazole antifungal Lecture 5 · slide 12; also Lecture 4 slide 31	ketoconazole Only 1 named in the lecture.	Alters fungal cell wall permeability by inhibiting CYP450 (cytochrome P450).	Systemic fungal infections.	None listed as such. Cautions drawn from the adverse effects: QT prolongation, hepatic toxicity, and CYP3A4 (cytochrome P450 3A4) inhibition, which raises levels of other drugs.	QTc (corrected QT interval) prolongation; hyperlipidemia; orthostatic hypotension; hepatitis, abnormal liver tests, cirrhosis, hepatic failure.	BLACK BOX FDA (Food and Drug Administration) boxed warning for oral ketoconazole: liver injury and QT prolongation with interacting drugs (the boxed warning itself is not stated in the lecture; hepatic failure is).
Nonabsorbable antifungal Lecture 5 · slide 13	nystatin Only 1 named in the lecture.	Binds sterols in the fungal cell membrane, increasing permeability.	Oral candidiasis (oral suspension), for example with inhaled steroids, human immunodeficiency virus infection or chemotherapy.	None listed.	Diarrhea; nausea; stomach pain; vomiting.	—

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Salicylates Lecture 5 · slides 15-18, 20	aspirin (acetylsalicylic acid) Only 1 named in the lecture.	Irreversible, noncompetitive inhibitor of platelets; nonselective for cyclooxygenase-1 and cyclooxygenase-2.	Antiplatelet at the lowest doses; antipyretic-analgesic (fever, pain) at higher doses; anti-inflammatory at the highest of these doses.	Bleeding disorders; pregnancy (very low doses may help preeclampsia); children with fever from a viral illness (chickenpox, influenza); do not combine with anticoagulants or other nonsteroidal anti-inflammatory drugs.	Bleeding; gastrointestinal upset, nausea; hypersensitivity; tinnitus. Overdose (salicylism): hyperventilation, alkalosis, then acidosis, shock, coma, death.	Reye syndrome (children under 15, mortality 50%); no aspirin for a child with fever from a viral illness.
NSAIDs (nonsteroidal anti-inflammatory drugs) Lecture 5 · slides 21-23	ibuprofen, naproxen Only 2 named in the lecture.	Ibuprofen reversibly inhibits cyclooxygenase-1 and cyclooxygenase-2, decreasing prostaglandin synthesis.	Relief of mild to moderate pain (ibuprofen: anti-inflammatory, analgesic, antipyretic). Naproxen (Aleve): longer half-life, less frequent dosing.	Allergy to the product; may exacerbate asthma; avoid under 6 months of age; history of stomach ulcers or perforation; renal dysfunction.	Gastric or duodenal ulcers, perforation, bleeding; edema and fluid retention; acute renal failure.	BLACK BOX Prescription NSAIDs carry an FDA (Food and Drug Administration) boxed warning for cardiovascular events and gastrointestinal bleeding (not stated in the lecture). Ibuprofen lowers ACE (angiotensin-converting enzyme) inhibitor effect and raises lithium and methotrexate levels.
Analgesic and antipyretic (acetaminophen) Lecture 5 · slides 24-25	acetaminophen Only 1 named in the lecture.	Not fully elucidated.	Pain; fever above 100 degrees Fahrenheit.	Allergy to the product (the only one listed). Alcohol increases the risk of liver damage (chronic use).	No notable side effects at therapeutic doses; very well tolerated.	Do not exceed 4 g in 24 hours. Alcohol plus acetaminophen raises the risk of liver damage.
First-generation H1 (histamine-1) antihistamines Lecture 5 · slides 31-37	chlorpheniramine, dimenhydrinate, diphenhydramine, hydroxyzine, meclizine, promethazine	Block histamine-1 receptors: block smooth muscle contraction and rapid vasodilation; decrease secretions, edema, hives and itch. Also antimuscarinic, antiemetic and antiserotonergic.	Allergic reactions (rhinitis, urticaria, insect bites, drug hypersensitivity); motion sickness, nausea and vestibular disturbances; over-the-counter sleep aids; adjunct in anaphylaxis.	Additive sedation with other central nervous system depressants or alcohol. Excitation instead of sedation at higher doses (especially in children) and in overdose.	Sedation (the major effect); gastrointestinal disturbances; dry mouth, urinary retention, blurred vision; dermatitis and photosensitivity with topical use.	BLACK BOX Sedation is the major side effect; promethazine has the strongest antimuscarinic action. Promethazine carries an FDA (Food and Drug Administration) boxed warning (tissue injury with injection; fatal breathing depression under age 2); not stated in the lecture.
Second-generation H1 (histamine-1) antihistamines Lecture 5 · slides 32, 35, 37	cetirizine, fexofenadine, loratadine	Block histamine-1 receptors but enter the central nervous system far less (much less sedation).	Allergic reactions (allergic rhinitis, urticaria, insect bites, drug hypersensitivity).	None listed.	Much less sedation: low with cetirizine, very low with fexofenadine and loratadine; very low anticholinergic effects.	Duration: cetirizine and loratadine 24 hours, fexofenadine 12 hours.
Intranasal antihistamine Lecture 5 · slides 38-39	azelastine Only 1 named in the lecture.	Histamine-1 (H1) receptor antagonist.	Allergic rhinitis and vasomotor rhinitis.	None listed. Do not use with other nasal sprays.	Bitter taste; epistaxis (nosebleed).	If a nosebleed develops, stop and follow up immediately. Spray away from the septum; do not tilt the head back.
Nasal corticosteroids Lecture 5 · slides 42-43	beclomethasone, budesonide, flunisolide, fluticasone, mometasone, triamcinolone	Not given in this lecture.	Allergic rhinitis and vasomotor rhinitis.	None listed; interactions: no information available.	Epistaxis; septal perforation; unpleasant taste.	Name trap: many have inhaled versions for asthma (Flonase is not Flovent).

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Systemic glucocorticoids Lecture 5 · slides 44, 46-48	dexamethasone, prednisone, prednisolone	Not given in this lecture (dexamethasone is classed as a synthetic adrenocortical steroid).	Allergic rhinitis, allergic conjunctivitis and drug hypersensitivity reactions.	Systemic fungal infections (dexamethasone); allergy to the product; infections, especially fungal (prednisone).	Sodium and fluid retention, heart failure, hypertension, low potassium; glucose intolerance; cushingoid features; tendon rupture, fractures; cataracts, glaucoma, raised eye pressure; weakened immunity.	Do not stop abruptly (dexamethasone after more than 1 week; prednisone: rebound symptoms). Avoid chickenpox or measles exposure and live vaccines.
Nasal decongestants (alpha agonists) Lecture 5 · slides 50-53	oxymetazoline, pseudoephedrine, phenylephrine	Alpha agonists that constrict blood vessels in the nasal mucosa (oxymetazoline, pseudoephedrine). Phenylephrine: mechanism not stated in the lecture.	Nasal congestion. Pseudoephedrine (oral) also treats congestion from the common cold, hay fever, allergies and sinus congestion, and eustachian tube dysfunction from viral infection.	None listed. Oxymetazoline interacts with monoamine oxidase inhibitors and antidepressants; pseudoephedrine decreases the effect of antihypertensives.	Oxymetazoline: rebound rhinitis (rhinitis medicamentosa) after more than 3 to 5 days; hypertension. Pseudoephedrine: tachycardia, hypertension, headache.	Oxymetazoline: no more than 3 to 5 days (rebound rhinitis).
Antitussives (cough suppressants) Lecture 5 · slides 57-58	benzonatate, dextromethorphan Only 2 named in the lecture.	Benzonatate anesthetizes the stretch receptors in the lungs. Dextromethorphan (related to codeine) suppresses the medullary cough center through sigma receptor activation.	Symptomatic relief of non-productive cough (benzonatate); cough (dextromethorphan).	Benzonatate: allergy to the product or related products (tetracaine). Dextromethorphan: taking, or within 2 weeks of taking, a monoamine oxidase inhibitor.	Benzonatate: local anesthesia from chewing the capsule. Dextromethorphan: confusion, excitement, agitation; serotonin syndrome risk with other pro-serotonergic drugs.	Dextromethorphan: never with a monoamine oxidase inhibitor (or within 2 weeks). Swallow benzonatate whole.
Expectorants and mucolytics Lecture 5 · slides 59, 61-63	guaifenesin, dornase alfa, hypertonic saline, N-acetylcysteine	Guaifenesin: loosens mucus, lowers viscosity. Dornase alfa cleaves DNA (deoxyribonucleic acid) from degenerating neutrophils, thinning cystic fibrosis sputum. N-acetylcysteine: splits disulfide bonds in mucoproteins. Hypertonic saline: hydrates the airway, aids clearance.	Guaifenesin: loosens mucus. Dornase alfa: mild to moderate cystic fibrosis lung disease (better lung function, fewer exacerbations). Inhaled N-acetylcysteine: mucolytic; traditionally used for acetaminophen toxicity.	Guaifenesin: allergy to the product; no interactions listed.	Guaifenesin: nausea, vomiting. Inhaled N-acetylcysteine: rotten-egg smell, nausea, vomiting, bronchospasm.	Inhaled N-acetylcysteine can cause bronchospasm.
ACE (angiotensin-converting enzyme) inhibitors Lecture 6 · slides 11-14, 16-22; also Lecture 8 slide 35	captopril, lisinopril, enalapril, benazepril, fosinopril, trandolapril, quinapril, ramipril, perindopril, moexipril	Block angiotensin-converting enzyme: less angiotensin II (less vasoconstriction, aldosterone, sodium reabsorption and norepinephrine release) and less bradykinin breakdown.	Hypertension (preferred in diabetics); heart failure and left ventricular dysfunction; after myocardial infarction (lower mortality); diabetic nephropathy; coronary artery disease with diabetes or left ventricular dysfunction.	Pregnancy (2nd and 3rd trimesters). Caution in kidney disease (may sharply lower filtration), with potassium-sparing diuretics, potassium supplements or salt substitutes (hyperkalemia); nonsteroidal anti-inflammatory drugs blunt the effect.	Dry cough (5-15%, bradykinin); hyperkalemia; first-dose hypotension; kidney impairment; angioedema (rare, 0.1-0.5%); fetal harm.	BLACK BOX FDA (Food and Drug Administration) boxed warning: fetal toxicity (birth defects, fetal death; contraindicated in the 2nd and 3rd trimesters). Angioedema: stop the drug.
ARBs (angiotensin receptor blockers) Lecture 6 · slides 25-30	candesartan, olmesartan, losartan, azilsartan, eprosartan, irbesartan, telmisartan, valsartan	Block AT1 (angiotensin II type 1) receptors with high affinity and slow dissociation: vasodilation, less aldosterone, sodium reabsorption and norepinephrine release. They do not inhibit bradykinin breakdown.	Hypertension (often with a diuretic); left ventricular dysfunction when an ACE (angiotensin-converting enzyme) inhibitor is not tolerated; diabetic nephropathy.	Pregnancy (not given in the 2nd and 3rd trimesters). Caution: hyperkalemia with kidney disease or potassium-sparing diuretics; first-dose hypotension; kidney impairment.	Hyperkalemia; hypotension; kidney impairment; fetal harm. No cough and no bradykinin effect; lower angioedema rate.	BLACK BOX FDA (Food and Drug Administration) boxed warning: fetal toxicity (not given in the 2nd and 3rd trimesters). In heart failure start low and titrate up.

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Non-dihydropyridine calcium channel blockers Lecture 6 · slides 33, 40-42, 44-47, 75	diltiazem, verapamil Only 2 named in the lecture.	Block L-type calcium channels in vascular smooth muscle, heart muscle and the sinoatrial and atrioventricular nodes; reduce the slow inward current AND slow its recovery, so atrioventricular conduction slows.	Angina; hypertension; supraventricular tachycardia; atrial fibrillation or flutter. Also listed for calcium channel blockers as a class: diastolic heart failure, migraine prophylaxis.	Advanced heart block; hypotension. Relative: heart failure, liver disease, gastroesophageal reflux disease.	Flushing, headache, hypotension, peripheral edema; negative inotropy (bradycardia, first-degree block, worse heart failure); nausea, constipation; raised liver tests; gingival hyperplasia.	Inhibit CYP3A4 (cytochrome P450 3A4) and P-glycoprotein: raise levels of statins, digoxin, cyclosporine, tacrolimus. Amiodarone: slower sinus rate or worse atrioventricular block. Beta blockers: additive drops in pressure, rate and contractility.
Dihydropyridine calcium channel blockers Lecture 6 · slides 38-40, 49, 51-52	amlodipine, nifedipine, nicardipine, felodipine, isradipine, nisoldipine, nimodipine	Block L-type calcium channels mainly in arterial smooth muscle (3 to 10 times more sensitive than heart muscle): vasodilation and lower afterload, with no effect on preload, veins or atrioventricular conduction.	Amlodipine, nifedipine, nicardipine: angina and hypertension. Felodipine, isradipine, nisoldipine: hypertension. Nimodipine: subarachnoid hemorrhage.	Severe aortic stenosis; unstable angina or recent myocardial infarction (immediate-release form). Avoid short-acting formulations.	Peripheral edema; dyspnea, wheezing; rebound tachycardia; gastrointestinal and central nervous system effects; skin reactions; gynecomastia; gingival hyperplasia.	Avoid short-acting formulations. Immediate-release forms are contraindicated in unstable angina or recent myocardial infarction.
Non-selective beta blockers (first generation) Lecture 6 · slides 56-57, 60, 63, 65-69, 71-74	propranolol, nadolol, timolol, pindolol, penbutolol, sotalol	Block cardiac beta-1 receptors (lower cardiac output) and renin release (less angiotensin II); block beta-2 receptors (bronchospasm) and presynaptic beta-2 receptors (less norepinephrine); lower central sympathetic activity.	Hypertension (not first line; best in young patients with fast heart rate, high catecholamines or renin); angina; after myocardial infarction; supraventricular arrhythmias; migraine prophylaxis (propranolol, timolol); hyperthyroid symptoms; essential tremor.	Bronchospasm or asthma (non-selective agents are worst); heart block; slow heart rate. Caution in diabetes (masks hypoglycemia), peripheral artery disease and heart failure.	Bronchoconstriction; fatigue, bradycardia, heart block; disturbed glucose control; cold extremities (Raynaud phenomenon); depression, nightmares (more with lipid-soluble propranolol, penbutolol); raised triglycerides.	BLACK BOX Sudden withdrawal causes acute angina, myocardial infarction and a marked blood pressure rise: taper slowly. FDA (Food and Drug Administration) boxed warning on propranolol, nadolol and timolol (not stated in the lecture). Sotalol: boxed warning for life-threatening arrhythmias.
Selective beta-1 blockers (second generation) Lecture 6 · slides 56, 61, 63, 67-68, 70-73	metoprolol, atenolol, bisoprolol, acebutolol, esmolol	Selectively block cardiac beta-1 receptors: lower heart rate and contractility (lower cardiac output) and lower renin release, so less angiotensin II.	Hypertension (not first line); angina; acute myocardial infarction (avoid agents with intrinsic sympathomimetic activity, such as acebutolol); heart failure (metoprolol succinate, bisoprolol); esmolol is intravenous.	Bronchospasm (less likely than with non-selective agents); heart block; bradycardia. Caution in diabetes and peripheral artery disease.	Fatigue, bradycardia, heart block; bronchoconstriction (less likely); disturbed glucose control that masks hypoglycemia; cold extremities.	BLACK BOX Sudden withdrawal: acute angina, myocardial infarction, marked blood pressure rise. FDA (Food and Drug Administration) boxed warning on atenolol and metoprolol (not stated in the lecture).
Third-generation beta blockers (vasodilating) Lecture 6 · slides 59, 62-63, 65, 70-74	carvedilol, labetalol, carteolol, betaxolol The slide groups carteolol and betaxolol (beta-1 selective) here too.	Beta blockade plus direct vasodilation: alpha-1 blockade (carvedilol, labetalol), beta-2 agonism, nitric oxide production, calcium entry blockade, potassium channel opening, antioxidant activity.	Hypertension; heart failure (carvedilol). Betaxolol, carteolol and timolol are also used for glaucoma.	As for all beta blockers: bronchospasm, heart block, bradycardia; caution in diabetes and peripheral artery disease.	Class effects: fatigue, bradycardia, heart block, bronchoconstriction, disturbed glucose control, cold extremities, central nervous system effects.	Sudden withdrawal syndrome applies to every beta blocker (acute angina, myocardial infarction, marked blood pressure rise): taper slowly.
Alpha-1 adrenergic blockers Lecture 6 · slides 77-84	prazosin, terazosin, doxazosin, tamsulosin	Reversible blockade of vascular alpha-1 receptors: dilates precapillary arterioles and lowers total peripheral resistance, with a reflex rise in heart rate. Tamsulosin is alpha-1a selective with little effect on blood vessels.	Hypertension (prazosin with beta blockers and diuretics; mildly improves lipids); benign prostatic hyperplasia (terazosin, doxazosin, tamsulosin). Terazosin and doxazosin have longer half-lives (once daily).	None listed. Caution in cardiac and renal failure; nonsteroidal anti-inflammatory drugs blunt the response; beta blockers may worsen postural hypotension.	Orthostatic hypotension, postural dizziness, headache, drowsiness, lack of energy; mild reflex tachycardia; sodium and water retention; impotence.	Orthostatic hypotension is the main side effect (postural dizziness).

PHARMACOLOGY DRUG STUDY CHART - Pharmacology I Exam 2

Class of Drugs	Generic Names (3+ Drugs)	Mechanism of Action (MOA)	Indications	Contraindications	Adverse Effects	BLACK BOX WARNING / Special Indications or Contraindications
Central sympatholytics (alpha-2 agonists) Lecture 6 · slides 75, 86-87, 89-95	clonidine, guanfacine Only 2 named in the lecture.	Stimulate postsynaptic alpha-2 receptors in the brainstem (nucleus of the solitary tract, ventral lateral medulla), lowering sympathetic outflow, peripheral resistance, heart rate and cardiac output; imidazoline receptors may play a role.	Hypertension: suitable for monotherapy; works well in the elderly; efficacy independent of age, race and gender; no negative effects on lipids. Clonidine also blunts opiate withdrawal.	None listed. Narrow therapeutic range; clonidine causes sodium retention, so it is often given with a diuretic.	Drowsiness or sedation; dry mouth; sexual dysfunction; orthostatic hypotension, bradycardia; clonidine raises blood glucose.	Abrupt withdrawal causes severe rebound hypertension: never stop suddenly. Clonidine plus a beta blocker lowers blood pressure further.
Direct vasodilators Lecture 6 · slides 97, 99-107	hydralazine, minoxidil, nitroprusside	Direct arteriolar dilation (lower peripheral resistance) with reflex sympathetic activation. Hydralazine: unclear (more cyclic guanosine monophosphate, nitric oxide). Minoxidil: opens potassium channels (hyperpolarization). Nitroprusside: releases nitric oxide; dilates veins and arterioles.	Hydralazine: chronic hypertension (with a diuretic and beta blocker). Minoxidil: severe or refractory hypertension (triple therapy). Nitroprusside (intravenous infusion): hypertensive crisis.	Hydralazine: coronary artery disease, ischemia, elderly. Minoxidil: none listed (its ischemia and arrhythmias are adverse effects).	Reflex tachycardia, fluid retention; headache, flushing; hydralazine lupus syndrome; minoxidil hypertrichosis, ischemia, arrhythmias; nitroprusside cyanide toxicity (sodium thiosulfate) and thiocyanate toxicity.	BLACK BOX FDA (Food and Drug Administration) boxed warnings: minoxidil (pericardial effusion, worsening angina; give with a beta blocker and diuretic) and nitroprusside (cyanide toxicity) (not stated in the lecture, except the cyanide toxicity).
HMG-CoA (3-hydroxy-3-methylglutaryl coenzyme A) reductase inhibitors (statins) Lecture 7 · slides 15, 18, 21-25, 65	atorvastatin, rosuvastatin, simvastatin, pravastatin, lovastatin, fluvastatin, pitavastatin	Reduce hepatic cholesterol synthesis, lowering intracellular cholesterol, which upregulates the LDL (low-density lipoprotein) receptor and increases uptake of non-HDL (high-density lipoprotein) particles from the blood.	First-line when low-density lipoprotein (LDL) cholesterol lowering is indicated; most efficacious and best tolerated. Benefit groups: existing cardiovascular disease; LDL above 190 mg/dL; diabetes aged 40-75 with LDL 70-189; no disease or diabetes, LDL 70-189 and 10-year risk above 7.5%.	Liver disease; pregnancy (current labeling weighs risk and benefit). Relative: cyclosporine or other immunosuppressants, gemfibrozil, niacin, erythromycin.	Headache, sleep disturbance, fatigue, gastrointestinal upset, flu-like symptoms; raised liver enzymes (0.5-2.5%); myalgia and myopathy (0.2-0.4%); rare rhabdomyolysis.	Muscle toxicity: stop the statin. Use fibrates cautiously with statins. Interactions via CYP3A4 (cytochrome P450 3A4) inhibitors: verapamil, amiodarone, grapefruit juice; also niacin and fibrates.
Cholesterol absorption inhibitor Lecture 7 · slides 28-31	ezetimibe Only 1 named in the lecture.	Selectively inhibits intestinal cholesterol absorption: less cholesterol reaches the liver, so more hepatic LDL (low-density lipoprotein) receptors and less cholesterol in atherogenic particles.	Lowers low-density lipoprotein (LDL) cholesterol; adds another 15-20% of LDL lowering when added to a statin. Taken with or without meals.	None listed. Interactions: fibrates (gallstones, myopathy); bile acid sequestrants and antacids lower levels; cyclosporine raises levels.	Gastrointestinal effects; raised liver transaminases when given with statins.	Fibrates with ezetimibe raise the risk of cholelithiasis (gallstones) and myopathy.
Fibrates (fibric acid derivatives) Lecture 7 · slides 15, 33-39	gemfibrozil, fenofibrate, bezafibrate	Activate PPAR-alpha (peroxisome proliferator-activated receptor alpha): more fatty acid oxidation, so less secretion of triglyceride-rich VLDL (very-low-density lipoprotein); more apolipoprotein A-I, raising HDL (high-density lipoprotein).	Primary indication: triglycerides above 1000 mg/dL or low high-density lipoprotein (HDL). Lower triglycerides 20-50%, raise HDL 10-35%.	Pregnancy; severe liver or kidney dysfunction; existing gallbladder disease.	Nausea, abdominal pain, diarrhea; cholelithiasis (gallstones); myopathy.	Raise the anticoagulant effect of warfarin; myopathy risk with statins.
Bile acid sequestrants (resins) Lecture 7 · slides 15, 42, 46-49	cholestyramine, colestipol, colesevelam	Bind bile acids in the gut, preventing enterohepatic recirculation; the liver makes more bile acids and more LDL (low-density lipoprotein) receptors, lowering LDL. Not absorbed.	Mainly combined with a statin, or when only a modest low-density lipoprotein (LDL) reduction is needed. The safest (no systemic effects); approved for children, adolescents and pregnancy.	Absolute: familial dysbetalipoproteinemia; triglycerides above 400 mg/dL. Relative: triglycerides above 200 mg/dL.	Bloating, flatulence, fullness, constipation, nausea; malabsorption of vitamins A, D, E, K and folic acid; may raise VLDL (very-low-density lipoprotein) and triglycerides.	Bind other drugs (digoxin, warfarin, thyroxine, beta blockers, thiazides); give the drug 1 hour before or 4 hours after the resin.

PHARMACOLOGY DRUG STUDY CHART - Pharmacology I Exam 2

Class of Drugs	Generic Names (3+ Drugs)	Mechanism of Action (MOA)	Indications	Contraindications	Adverse Effects	BLACK BOX WARNING / Special Indications or Contraindications
Niacin (nicotinic acid) Lecture 7 · slides 50-51, 53, 55-56, 58, 61	niacin (nicotinic acid), niacin immediate release (Niacor), niacin extended release (Niaspan), inositol hexaniacinate	Decreases hepatic VLDL (very-low-density lipoprotein) and apolipoprotein B production, so less conversion to LDL (low-density lipoprotein); decreases mobilization of free fatty acids and triglyceride synthesis; raises HDL (high-density lipoprotein).	Atherogenic dyslipidemia, alone or combined when low-density lipoprotein (LDL) is also elevated. Lowers triglycerides 20-50% and raises high-density lipoprotein (HDL) 15-35%. Niacinamide is NOT effective.	Absolute: chronic liver disease. Relative: peptic ulcer disease, symptomatic gout, significant hyperuricemia, diabetes (glucose intolerance).	Cutaneous flushing (prostaglandin-mediated); nausea, abdominal discomfort; at larger doses raised liver tests, glucose and uric acid; decreased glucose tolerance.	Flushing is prostaglandin-mediated: premedicate with aspirin. Niacinamide (nicotinamide) does not lower lipids.
PCSK9 (proprotein convertase subtilisin/kexin type 9) inhibitors Lecture 7 · slide 60	alirocumab, evolocumab Only 2 named in the lecture.	Injectable monoclonal antibodies: PCSK9 (proprotein convertase subtilisin/kexin type 9) processes hepatic LDL (low-density lipoprotein) receptors, so blocking it keeps the receptors active longer and lowers LDL.	Lower low-density lipoprotein (LDL) cholesterol (drops of 43-58%).	None listed.	Hypersensitivity reactions are the most serious adverse reaction.	Injectable only and expensive.
Beta blockers (antianginal) Lecture 8 · slides 8, 13, 16-19, 39, 55	metoprolol, atenolol, propranolol, nadolol, carvedilol, labetalol	Lower myocardial oxygen demand: decrease heart rate, contractility and systolic wall tension. No effect on oxygen supply.	First-line for angina without contraindications, especially with hypertension, anxiety, supraventricular arrhythmias, heart failure or a prior myocardial infarction. In acute coronary syndrome: fewer deaths, smaller infarcts, less sudden cardiac death.	Heart rate under 60; systolic pressure under 100; atrioventricular block; acute decompensated heart failure. Precautions: reactive airway disease, systolic heart failure, diabetes, peripheral vascular disease. Avoid in variant (Prinzmetal) angina.	Hypotension, bradycardia, hyperglycemia, dyslipidemia; fatigue, sexual dysfunction, nightmares, worsened claudication.	BLACK BOX Avoid rapid discontinuation. FDA (Food and Drug Administration) boxed warning for abrupt cessation: metoprolol, atenolol, propranolol, nadolol (not stated in the lecture). Avoid in variant angina.
Calcium channel blockers (antianginal) Lecture 8 · slides 20-24, 34	diltiazem, verapamil, nifedipine, amlodipine, felodipine	Lower demand (lower systolic wall tension; non-dihydropyridines also slow the heart and lower contractility) and raise supply (mild dilation at fixed stenoses, relief of vasospasm).	Non-dihydropyridines: initial therapy when beta blockers are contraindicated or not tolerated. Dihydropyridines: added to a beta blocker when it alone fails. Chosen for vasospastic angina, severe peripheral vascular disease, asthma, uncontrolled diabetes.	Systolic pressure under 100. Non-dihydropyridines: heart rate under 60, acute heart failure, ejection fraction under 40%, atrioventricular block. Precaution: a non-dihydropyridine with a beta blocker; CYP3A4 (cytochrome P450 3A4) interactions.	Hypotension; dihydropyridines: headache, flushing, peripheral edema. Counsel on dizziness and constipation.	Avoid short-acting agents (nifedipine). With left ventricular dysfunction, amlodipine is the only calcium channel blocker used.
Nitrates (short- and long-acting) Lecture 8 · slides 25-27, 29-33, 38, 54	nitroglycerin, isosorbide mononitrate, isosorbide dinitrate	Nitric oxide raises cyclic guanosine monophosphate and lowers cytosolic calcium: smooth muscle relaxation and vasodilation (lower left ventricular volume). Also dilate coronary arteries, relieve vasospasm, and have antiplatelet effects.	Sublingual tablet or spray: relieve acute ischemia, prevent effort-induced angina; call for help if no relief 5 minutes after the first dose. Long-acting forms: adjunct, not monotherapy. In acute coronary syndrome: pain relief only, no mortality benefit.	Phosphodiesterase type 5 inhibitors (sildenafil, tadalafil, vardenafil); aortic valve stenosis; obstructive cardiomyopathy; hypotension.	Headache, flushing, postural hypotension, reflex tachycardia; tachyphylaxis (tolerance), managed with a nitrate-free interval.	NEVER with sildenafil, tadalafil or vardenafil: hypotension, myocardial infarction or stroke.
Antiplatelets Lecture 8 · slides 13, 37, 52-53; also Lecture 5 slide 15; Lecture 5 slide 16, Lecture 5 slide 17	aspirin, clopidogrel Only 2 named in the lecture.	Aspirin: irreversible platelet inhibition (nonselective cyclooxygenase inhibitor). Clopidogrel: mechanism not given. Both increase coronary blood flow.	Aspirin for all patients with ischemic heart disease, to prevent acute coronary syndrome; chew it at the first sign of chest pain (fewer deaths and reinfarctions). Clopidogrel: as effective as aspirin in secondary prevention; used for aspirin allergy.	Aspirin: allergy; recent gastrointestinal bleeding; recent intracranial hemorrhage; bleeding disorders.	Bleeding; gastrointestinal upset (aspirin).	BLACK BOX Clopidogrel: FDA (Food and Drug Administration) boxed warning for reduced effect in CYP2C19 (cytochrome P450 2C19) poor metabolizers (not stated in the lecture). Aspirin: never for a child with viral fever (Reye syndrome).

PHARMACOLOGY DRUG STUDY CHART - Pharmacology I Exam 2

Class of Drugs	Generic Names (3+ Drugs)	Mechanism of Action (MOA)	Indications	Contraindications	Adverse Effects	BLACK BOX WARNING / Special Indications or Contraindications
Fibrinolytics (thrombolytics) Lecture 8 · slides 58-59, 61-65, 68	streptokinase, alteplase, reteplase, tenecteplase	Activate conversion of plasminogen to plasmin, which lyses fibrin. Streptokinase forms a 1:1 complex with plasminogen; alteplase, reteplase and tenecteplase act only on fibrin-bound plasminogen.	ST-elevation myocardial infarction: under 75 years and within 12 hours of symptom onset. Less benefit over 75 years, after 6 hours, or in non-ST-elevation myocardial infarction. Relatively few patients receive them.	Active bleeding or bleeding disorder; surgery within 10 days; serious gastrointestinal bleeding within 3 months; severe hypertension (diastolic above 110 mmHg); prior stroke or intracranial tumor; aortic dissection; pericarditis; prior streptokinase; pregnancy.	Bleeding (0.5-7%; intracranial hemorrhage 0.4-0.94%); allergic reactions, fever, chills, rash (mainly streptokinase, urokinase); anaphylaxis; ventricular arrhythmias.	Not recommended in non-ST-elevation acute coronary syndrome: bleeding risk exceeds benefit.
Opioid analgesic Lecture 8 · slide 56	morphine Only 1 named in the lecture.	Not given in this lecture.	Chest pain unresponsive to nitrates; used in ST-elevation myocardial infarction.	Not listed as a contraindication. Caution: may increase mortality in unstable angina and non-ST-elevation myocardial infarction; use is controversial.	Hypotension; allergy.	BLACK BOX Opioids carry an FDA (Food and Drug Administration) boxed warning (addiction, life-threatening respiratory depression); not stated in the lecture. May raise mortality in unstable angina or non-ST-elevation myocardial infarction.
Other acute coronary syndrome antithrombotics (heparins, glycoprotein IIb/IIIa inhibitors, P2Y₁₂ (platelet adenosine diphosphate receptor) antagonists) Lecture 8 · slides 66, 68	enoxaparin, heparin Only enoxaparin is named; the other agents appear as class names.	Not given in this lecture.	Heparins: used with fibrinolysis or antiplatelet agents; enoxaparin preferred over heparin in non-ST-elevation acute coronary syndrome. P2Y ₁₂ antagonists: before percutaneous coronary intervention; clot prophylaxis with stents. Glycoprotein IIb/IIIa inhibitors: not routine before intervention.	None listed.	Not given in this lecture.	BLACK BOX Enoxaparin carries an FDA (Food and Drug Administration) boxed warning: spinal or epidural hematoma with neuraxial anesthesia or puncture (not stated in the lecture).